

Leonardo Maglanoc

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Profile

Robot Learning & AI Engineer with an M.Sc. from TUM. Experience spans dexterous manipulation at Foundation Robotics, neuro-inspired ML at BMW Research, safe human-robot interaction at TUM, and 3D perception/SLAM. Born in Norway and raised in Germany, I'm particularly excited to contribute to Europe's deep-tech ecosystem.

Experience

Master's Thesis Collaboration | 10/2025 – 05/2026

Foundation Robotics & TUM MIRMI, Munich, full-time

- Developed Lang2Tool: mapped language instructions to validated LLM tool calls and parameterized SE(3) tool trajectories.
- Integrated trajectory goals with a learned low-level policy for dexterous tool manipulation in simulation.

AI/ML Research Intern | 02/2025 – 08/2025

BMW Group Research, Munich, full-time

- Co-developed a Python/PyTorch simulator for oscillatory neural networks (ONNs) using Neural ODEs, in collaboration with IBM.
- Led a tactile-data study with 10 participants; trained and evaluated ONNs for a human-vehicle interaction demonstrator.

Robotics Research Assistant | 04/2023 – 05/2024

Technical University of Munich (TUM), part-time

- Contributed to SaRA-Shield: reachability-based runtime safety for formally verified human-robot interaction.
- Implemented kinematics and trajectory-planning algorithms; co-authored an IEEE Transactions on Robotics paper.

Computer Science Teaching Assistant – TUM | 11/2020 – 04/2021

- Taught Java and algorithmic problem-solving; graded assignments and examinations.

Selected Projects

SO-101 Robot Learning | 08/2026

- Built a LeRobot pipeline on a physical SO-101: teleoperation, dataset collection, ACT/SmolVLA training and real-time pick-and-place deployment.

LiDAR-Visual-Inertial SLAM – Guided Research, TUM | 04/2024 – 09/2024

- Implemented and evaluated LiDAR feature-extraction and voxel-filtering methods in C++ within a dense multi-sensor SLAM system using real HILTI 2022 sensor data.

Education

M.Sc. Robotics, Cognition, Intelligence – TUM | 04/2023 – 06/2026

Focus: robotics, computer vision and machine learning.

B.Sc. Informatics – TUM | 09/2019 – 03/2023

Thesis: *Provably Safe Human-Robot Interaction for Manipulation using Power and Force Limiting.*

Publication

J. Thumm, J. Balletshofer, **L. Maglanoc**, L. Muschal, M. Althoff. "A general safety framework for autonomous manipulation in human environments." *IEEE Transactions on Robotics*, 2026.

Technical Skills

Python · PyTorch · C++ · ROS2 · LeRobot · Linux · Git · Docker

Languages

German and Norwegian (native); English (C2); French (B1).